Multimodal Fashion Retrieval System

System Progress & Comprehensive Evaluation Report | Attribute Re-ranking vs Baseline

Target Index: 30,000 Fashionpedia Items	**Candidate Pool:** Top-100 FAISS Candidates	**Evaluation Date:** October 2026
Embedding Model: CLIP ViT-B/32	**Re-ranking Engine:** Garment Segment Mask + Color Matching	**Status:** Production Benchmark Passed

1. Executive Summary & Key Achievements

This progress report evaluates the end-to-end performance of the **Multimodal Fashion Search & Attribute Re-ranking Engine** against a standard baseline (pure CLIP dense cosine retrieval over FAISS). Standard dense multimodal models suffer from well-documented *attribute binding conflation* (e.g., misidentifying color assignments across co-occurring garments). Our fine-grained color/garment attribute re-ranker acts on top candidates, re-scoring by verified localized masks.

Key Result Highlights:
- ***Precision@10:*** *Skyrocketed from* ***46.0%*** *(Baseline) to* ***98.0%*** *(+113% relative gain).*
- ***MRR (Mean Reciprocal Rank):*** *Jumped from* ***0.5575*** *to* ***1.0000*** *(a top relevant item is now consistently ranked #1).*
- ***NDCG@10:*** *Improved from* ***0.4512*** *to* ***0.9864*** *(+118.6% ranking quality improvement).*
- ***Complex Queries:*** *Successfully untangles multi-garment bindings (e.g., 'white shirt with red tie', 'blue shirt and black pants') with 100% Top-10 precision.*

2. Aggregate Evaluation Metrics: Baseline vs Re-ranked

Metric	Text Baseline	Text Re-ranked	Text Gain	Image Baseline	Image Re-ranked	Image Gain
P@1	0.35	**1.00**	++65%	0.55	**1.00**	++45%
P@5	0.49	**1.00**	++51%	0.46	**1.00**	++54%
P@10	0.46	**0.98**	++52%	0.43	**0.90**	++46%
MRR	0.5575	**1.0000**	+79.4%	0.7042	**1.0000**	+42.0%
NDCG@10	0.4512	**0.9864**	+118.6%	0.4534	**0.9364**	+106.5%
MAP	0.0092	**0.0403**	+338.0%	0.0006	**0.0040**	+566.7%
HitRate@10	0.90	**1.00**	++10%	0.90	**1.00**	++10%

3. Benchmark Queries & Per-Query Retrieval Results

Below are all **20 evaluation queries** executed against the 30,000 item Fashionpedia FAISS index. Each query is categorized by complexity type. We compare **Re-ranked vs. Baseline** precision and ranking scores.

ID	Query Text	Type / Category	Pool Rel	Base P@10	Rerank P@10	Base NDCG	Rerank NDCG
TQ01	*'blue shirt'*	Single Attr	1155	0.20	**1.00**	0.305	**1.000**
TQ02	*'red dress'*	Single Attr	995	1.00	**1.00**	1.000	**1.000**
TQ03	*'black pants'*	Single Attr	3123	0.60	**1.00**	0.498	**1.000**
TQ04	*'white jacket'*	Single Attr	201	0.40	**1.00**	0.389	**1.000**
TQ05	*'green coat'*	Single Attr	51	0.50	**1.00**	0.498	**1.000**
TQ06	*'yellow skirt'*	Single Attr	88	0.40	**1.00**	0.520	**1.000**
TQ07	*'pink sweater'*	Single Attr	63	0.40	**1.00**	0.325	**1.000**
TQ08	*'gray hat'*	Single Attr	362	0.20	**1.00**	0.183	**1.000**
TQ09	*'brown bag'*	Single Attr	601	0.00	**1.00**	0.000	**1.000**
TQ10	*'orange shoe'*	Single Attr	1370	0.20	**1.00**	0.164	**1.000**
TQ11	*'blue shirt and black pants'*	Multi-Garment	4062	0.80	**1.00**	0.848	**1.000**
TQ12	*'white shirt with red tie'*	Multi-Garment	652	0.40	**1.00**	0.347	**1.000**
TQ13	*'red dress with black belt'*	Multi-Garment	1947	0.90	**1.00**	0.905	**1.000**
TQ14	*'green jacket and blue pants'*	Multi-Garment	3078	0.50	**1.00**	0.458	**1.000**
TQ15	*'pink skirt and white top'*	Multi-Garment	683	0.30	**1.00**	0.222	**1.000**
TQ16	*'someone wearing a bright yellow raincoat'*	Natural Lang	52	0.00	**0.60**	0.000	**0.727**
TQ17	*'casual blue jeans outfit'*	Natural Lang	2982	0.40	**1.00**	0.473	**1.000**
TQ18	*'woman in a red gown'*	Natural Lang	995	0.90	**1.00**	0.931	**1.000**
TQ19	*'man in a black blazer'*	Natural Lang	899	0.60	**1.00**	0.490	**1.000**
TQ20	*'summer dress in white'*	Natural Lang	300	0.50	**1.00**	0.467	**1.000**

Observation on TQ16 ('someone wearing a bright yellow raincoat'): *P@10 reached 0.60 (vs 0.00 in baseline). The ground truth for this natural language query has fewer matched items (52 relevant in total index), and our segmenter re-ranker accurately retrieved 6 verified yellow coats in top 10 while pure CLIP failed completely.*

4. Extended Question Bank for Text Evaluation

To systematically stress-test fashion text search engines beyond standard canonical queries, the following expanded test suites target critical linguistic and compositional retrieval challenges. These test categories can be incorporated into upcoming benchmark iterations.

A. Compositional & Attribute Binding Queries (Binding Problem)

• **TQ-EXT-01:** *'striped navy blue blazer with brass buttons'* — Tests multi-attribute binding (pattern + color + sub-component) to prevent returning plain navy suits.

• **TQ-EXT-02:** *'white crop top, olive green cargo pants, and white sneakers'* — Tests 3-way garment-color binding across upper-body, lower-body, and footwear simultaneously.

• **TQ-EXT-03:** *'burgundy hoodie inside a black denim jacket'* — Evaluates nested garment layering disambiguation.

• **TQ-EXT-04:** *'black shirt with white polka dots vs white shirt with black polka dots'* — Pairwise symmetry test for foreground/background color inversion.

B. Negative Attributes & Exclusion Testing

• **TQ-EXT-05:** *'formal cocktail dress without sequins or glitter'* — Checks if the system suppresses garments with noisy texture features.

• **TQ-EXT-06:** *'long sleeve cotton t-shirt without front graphics or print'* — Tests negation handling in aesthetic classification.

• **TQ-EXT-07:** *'casual summer shorts, no denim'* — Tests exclusion of dominating material classes in visual embeddings.

C. Subtle Color Shades & Material/Texture Nuance

• **TQ-EXT-08:** *'champagne satin slip dress vs beige linen dress'* — Differentiates subtle color gradients coupled with sheen/fabric physics.

• **TQ-EXT-09:** *'charcoal gray wool overcoat'* — Differentiates dark neutral shades (charcoal vs pure black vs heather gray).

• **TQ-EXT-10:** *'emerald green velvet blazer'* — Cross-evaluates gemstone color tone with reflective textile texture.

D. Occasion, Style Vibe & Contextual Abstraction

• **TQ-EXT-11:** *'business casual outfit for a summer tech conference'* — Evaluates open-domain semantic reasoning to retrieve cohesive garments.

• **TQ-EXT-12:** *'vintage 90s streetwear baggy denim outfit'* — Assesses aesthetic era and silhouette retrieval.

• **TQ-EXT-13:** *'minimalist monochrome monochrome athleisure outfit'* — Tests cohesive styling rules across a single color family.

E. Size, Silhouette & Fit Variations

• **TQ-EXT-14:** *'oversized chunky knit sweater with skinny jeans'* — Tests contrasting silhouette retrieval across co-occurring garments.

• **TQ-EXT-15:** *'high-waisted pleated wide-leg trousers'* — Assesses fine-grained structural cuts that CLIP embeddings frequently overlook.

5. Technical Next Steps & Optimization Roadmap

1. Pre-caching SegFormer Masks: Move on-the-fly garment segmentation inference to offline index pipelines for candidate pools > 100.

2. LLM-based Query Expansion: Incorporate few-shot query rewrites to support negative constraints ('without denim') before FAISS candidate retrieval.

3. Multi-modal Re-ranking Fusion: Weight localized garment color scores with dense text-image cosine scores using learned logistic regression weights.

4. Expanded Benchmark Suite: Integrate the 15 extended test queries above into automated CI regression testing.